Database Management Systems

Summary: Week 1 - 4

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Week 1 Summary

Comparison

Parameter	File Handling via Python	DBMS
Scalability with respect to amount of data	Very difficult to handle insert, update and querying of records	In-built features to provide high scalability for a large number of records
Scalability with respect to changes in structure	Extremely difficult to change the structure of records as in the case of adding or removing attributes	Adding or removing attributes can be done seamlessly using simple SQL queries
Time of execution	In seconds	In milliseconds
Persistence	Data processed using temporary data structures have to be manually updated to the file	Data persistence is ensured via automatic, system induced mechanisms
Robustness	Ensuring robustness of data has to be done manually	Backup, recovery and restore need minimum manual intervention
Security	Difficult to implement in Python (Security at OS level)	User-specific access at database level
Programmer's produc-	Most file access operations involve	Standard and simple built-in queries

Levels of Abstraction

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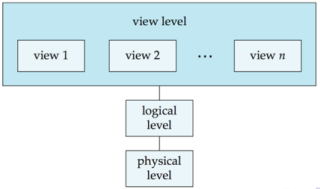

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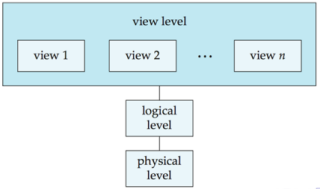

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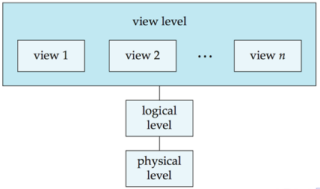
- **View level**: Application programs hide details of data types; it also hides information for security purposes (user interfaces)



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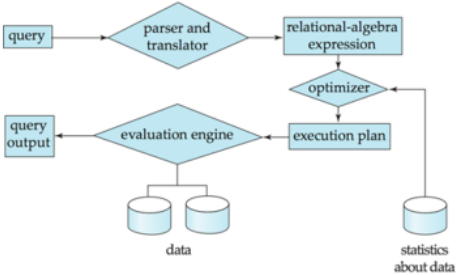

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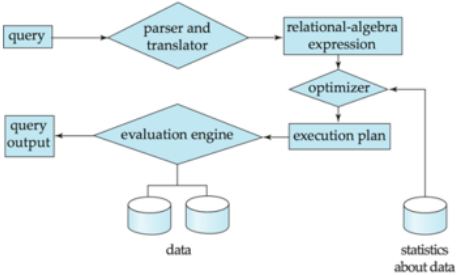

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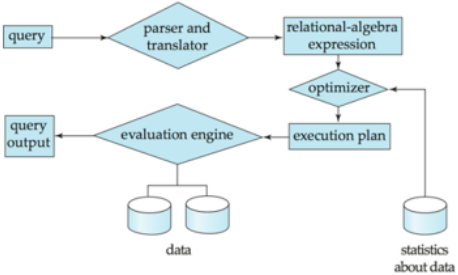

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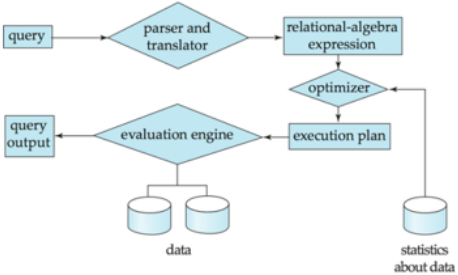

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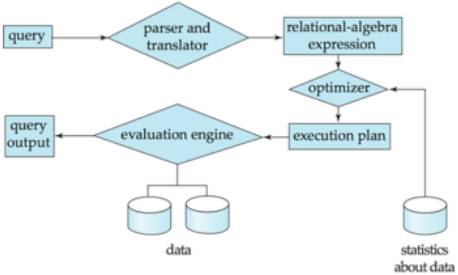

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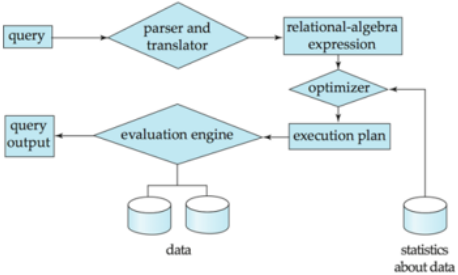
  - Parsing and translation → Optimization → Evaluation



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  - Transaction-management component - ensures that the database remains in a consistent state despite system failures and transaction failures.
  - Concurrency-control manager - controls the interaction among the concurrent transactions

Week 2 Summary

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- A **surrogate key** (or synthetic key) in a database is a unique identifier for either an entity in the modeled world or an object in the database.

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  - **Declarative programming** : requires a more descriptive style. The programmer must know what relationships hold between various entities

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- **Natural Join** : natural join operation on two relations matches tuples whose values are the same on all attribute names that are common to both relations. i.e. $r \bowtie s$

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  - **From clause** : lists the relations involved in the query.

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- **Order By** : We may specify `desc` for descending order or `asc` for ascending order, for each attribute; ascending order is the default. (i.e. `order by name desc`)

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- **IN operator** : The `in` operator allows you to specify multiple values in a where clause.
- **Duplicates** : In relations with duplicates, SQL can define how many copies of tuples appear in the result. Multiset versions of some relational algebra operators.

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  - Returns the name and age of students with a CGPA above 8.

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  - Returns the name and age of students having marks above 75.
- **Note**: You have to mention the domain variables in the same order as given in the table.

Example Problem Q1

Relation Students

Name	Age	Marks	Subject
David	23	78	Maths
Matthew	29	54	English
Anand	22	89	JAVA
Mitchel	21	56	Maths
Shaun	26	92	Maths

Relation Activity

Name	Sports	Awards	Points
David	Cricket	2	67
Matthew	Football	4	90
Anand	Cricket	5	80
Mitchel	Tennis	8	70
Shaun	Hockey	3	75

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Q). Return the names of students whose age is greater than 25, or who are enrolled in Maths.

- **RA**: $\Pi_{\text{Name}}(\sigma_{\text{age}>25 \vee \text{subject}=\text{'Maths'}}(\text{Students}))$
- **TRC**: $\{t \mid \exists\ s \in \text{students}(s.\text{age} > 25 \vee s.\text{subject} = \text{'Maths'} \wedge t.\text{name} = s.\text{name})\}$

Example Problem Q2

Relation Students

Name	Age	Marks	Subject
David	23	78	Maths
Matthew	29	54	English
...	...	...	...

Relation Activity

Name	Sports	Awards	Points
David	Cricket	2	67
Matthew	Football	4	90
...	...	...	...

Example Problem Q2

Relation Students

Name	Age	Marks	Subject
David	23	78	Maths
Matthew	29	54	English
...	...	...	...

Relation Activity

Name	Sports	Awards	Points
David	Cricket	2	67
Matthew	Football	4	90
...	...	...	...

Q). Return the names of students along with sports, whose age is less than 25 and who have secured more than 75 marks.

- **RA**: $\Pi_{\text{Name,Sports}}(\sigma_{\text{age}<25 \wedge \text{Marks}>75}(\text{Students} \bowtie \text{Activity}))$
- **TRC**: $\{t \mid \exists\ s \in \text{students} \exists\ a \in \text{activity}(s.\text{name} = a.\text{name} \wedge s.\text{age} < 25 \wedge s.\text{marks} > 75 \wedge t.\text{name} = s.\text{name} \wedge t.\text{sports} = a.\text{sports})\}$
- **DRC**: $\{\langle a, f\rangle \mid \exists\ b, c, d(\langle a, b, c, d\rangle \in \text{students} \wedge b < 25 \wedge c > 75) \wedge \exists\ e, g, h(\langle e, f, g, h\rangle \in$

E-R Diagram and Mapping Constraints

- Please go through the tutorials.

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many-to-many relationship

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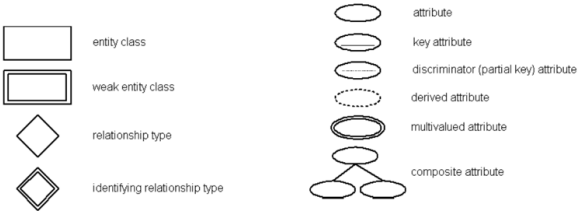
Total and Partial Participation

E-R diagrams symbols